

4/23

Crown

Name:

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REVIEW WK

1. How many molecules of N_2O_4 are in 66.3 g N_2O_4 ? The molar mass of N_2O_4 is 92.02 g/mol.A. 7.26×10^{23} N_2O_4 moleculesC. 4.59×10^{25} N_2O_4 moleculesB. 5.54×10^{25} N_2O_4 moleculesD. 4.34×10^{23} N_2O_4 moleculesat 2.01 M N_2O_4 , 6100.263, 3.673621, 4.3389623

2. Which of the following represent isotopes of lead (Pb)?

$^{204}_{82}\text{Pb}$	$^{206}_{82}\text{Pb}$	$^{204}_{81}\text{Pb}$	$^{207}_{83}\text{Pb}$	$^{208}_{82}\text{Pb}$
I	II	III	IV	V

A. III

B. I, II, and V

C. I and V

D. I, II, and IV

3. The bonding in CaCl_2 is best described as

A. each chlorine losing an electron, and calcium gaining two electrons

B. the sharing of electrons between calcium and chlorine atoms

C. calcium losing two electrons, and each chlorine gaining an electron

D. calcium losing an electron, and chlorine gaining an electron

4. Which of the following are strong electrolytes?

Hydrochloric acid (HCl)	Potassium chloride (KCl)	Sugar ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$)	Potassium hydroxide (KOH)
I.	II.	III.	IV.

A. I and II

B. II and III

C. I, II, and IV

D. All the above.

5. Determine the oxidation state of carbon in dichloromethane (CH_2Cl_2).

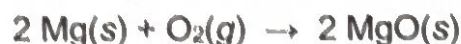
A. +4

B. 0

C. -1

D. +2

6. Magnesium burns in air with a dazzling brilliance to produce magnesium oxide:



When 5.25 g of magnesium burns, the theoretical yield of magnesium oxide is _____ g.

(Molar Mass MgO = 40.30g/mol)

A. 3.73

B. 4.15

C. 8.70

D. 16.6

7. Determine the limiting reactant (LR) when 13 mol of sodium and 6 mol of chlorine react according to the following equation:



A. LR= Na

B. LR= Cl_2 C. LR= Cl_2

D. LR= NaCl

8. How many moles of NaOH is required to reach the equivalence point of 1 L containing 20 moles of a HCl solution?

A. 2 moles of NaOH

B. 20 moles of NaOH

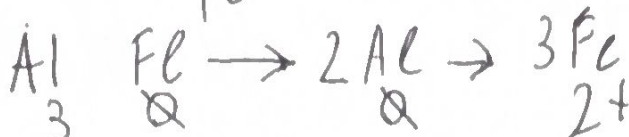
C. 10 moles of NaOH

D. 200 moles of NaOH

m/v

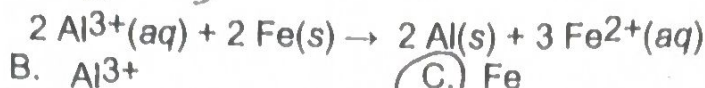
$$\frac{20 \text{ mol}}{1 \text{ L}} = 20 \text{ M}$$

Al: 3 → 0: Reduct: Oxidizing Agent
 Fe: 0 → 2: Oxidizt: Reducing Agent



9. What is undergoing oxidation?

A. Fe²⁺



B. Al³⁺

C. Fe

D. Al

10. Which of the following (with the specific heat capacity provided) would show the smallest temperature change upon **losing** 200.0 J of heat?

A. 25.0 g Hg, C_{Hg} = 0.14 J/g°C

C. 25.0 g Ag, C_{Ag} = 0.235 J/g°C

B. 25.0 g granite, C_{granite} = 0.79 J/g°C

11. Which of the following will cause the volume of an ideal gas to double in value?

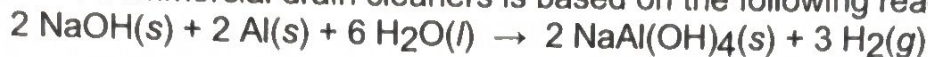
A. Lowering the absolute temperature by a factor of 2 at constant pressure.

B. Lowering the pressure by a factor of 2 while the temperature stays constant.

C. Lowering the absolute temperature by a factor of 2 while increasing the pressure by a factor of 2.

D. Raising the absolute temperature by a factor of 2 while decreasing the pressure by a factor of 2.

12. The action of some commercial drain cleaners is based on the following reaction:



What is the volume of H₂ gas formed at STP when 2.54 g of Al reacts with excess NaOH?

A. 3.16 L

B. 2.11 L

C. 4.76 L

D. 7.23 L

$$PV = nRT; V = (0.082058)(273.15)(0.141205248) = V = 3.16499\text{L}$$

$$(1.5)(2.54/26.98\text{L}) = \text{A} \text{ MH}_2$$

13. Identify the element that has a ground state electron configuration of [Ar]4s²3d¹⁰4p³.

A. Se

B. As

C. N

D. Sb

14. Give the numbers for *m_l* for a d orbital.

A. -2, -1, 0, +1, +2

B. 0, +1, +2

C. -1, 0, +1

D. +1, +2, +3

15. Of the following, which element has the **highest** first ionization energy?

A. Sn

B. Te

C. I

D. Kr

16. Use the bond energies provided to estimate ΔH°_{rxn} for the reaction below.



ΔH°_{rxn} = ?

Bond Bond Energy (kJ/mol)

C-C 347

C-H 414

C=C 611

C≡C 837

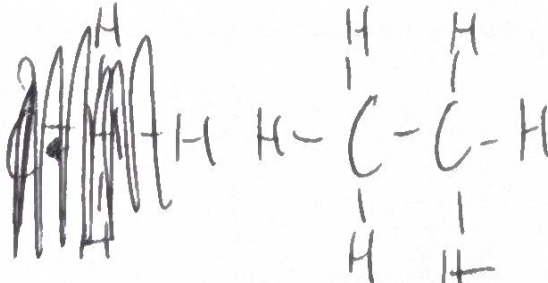
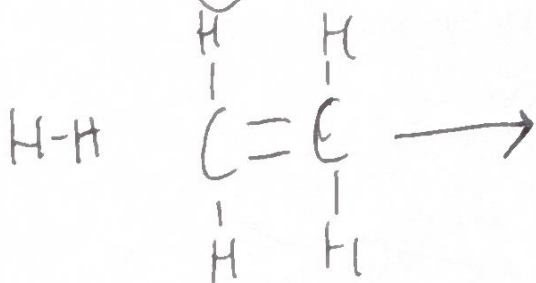
H-H 436

A. -128 kJ

B. +98 kJ

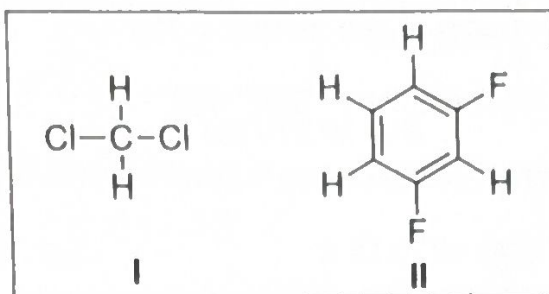
C. +700 kJ

D. -102 kJ



$$\begin{array}{rcl} \text{C}=\text{C} & 1 & -611 \\ \text{C}-\text{H} & 4 & -1656 \\ \text{H}-\text{H} & 1 & -436 \\ \hline \text{C}-\text{H} & 6 & -2484 \\ \text{C}-\text{C} & 1 & -347 \\ \hline & & -2831 \\ & & +128 \\ \hline & & -2703 \end{array}$$

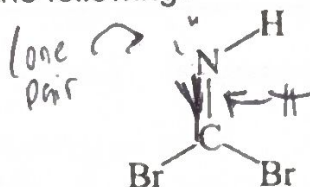
17. Determine the polarity of the dichloromethane and 1,3-difluorobenzene below:



- ☐ A. I = polar; II = nonpolar
☒ B. I = polar; II = polar

- C. I = nonpolar; II = nonpolar
 D. I = nonpolar; II = polar

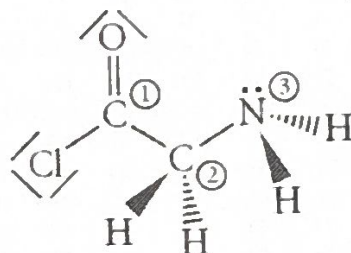
18. The following Lewis structure is incomplete. When completed, the Lewis structure of the following molecule will feature which of the following?



- ☒ A. At least one pi bond
☐ B. At least one lone pair of electrons on each atom

- ☐ C. Only nonpolar bonds
☒ D. More than four sigma bonds

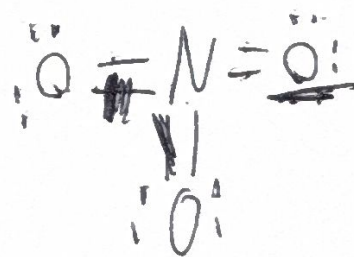
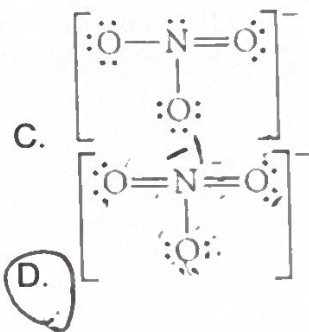
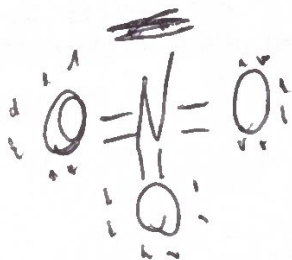
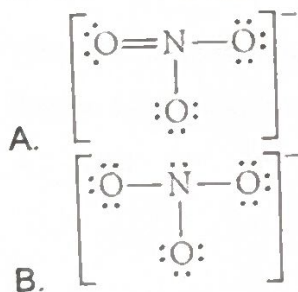
19. Consider the molecule below. Determine the molecular geometry at each of the 3 labeled atoms.



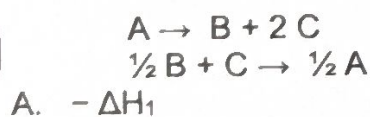
- ☒ A. 1 = trigonal planar, 2 = tetrahedral, 3 = trigonal pyramidal
☐ B. 1 = tetrahedral, 2 = tetrahedral, 3 = tetrahedral
☐ C. 1 = trigonal planar, 2 = tetrahedral, 3 = tetrahedral
☐ D. 1 = tetrahedral, 2 = tetrahedral, 3 = trigonal planar
☐ E. 1 = trigonal planar, 2 = trigonal pyramidal, 3 = trigonal pyramidal

I 10 ~~Tetrahedral~~ Trigonal planar
 II 40 Tetrahedral
 III 41 ~~Tetrahedral~~ Pyramidal
 Trigonal

20. Choose the **best** Lewis structure for NO_3^- .



21. For the generic reaction, determine the value of ΔH_2 in terms of ΔH_1 .



ΔH_1
 $\Delta H_2 = ?$
 B. $-2\Delta H_1$

C. $\frac{1}{2}\Delta H_1$

☒ D. $-\frac{1}{2}\Delta H_1$

$-\frac{1}{2}\Delta H_1$

Constants

Measurements of Pressure

$$1 \text{ atm} = 760 \text{ mmHg} = 760 \text{ Torr}$$

Gas Constants

$$R = 0.0821 \text{ L} \cdot \text{atm} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$$

Avogadro Constant

$$N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$$

Temperature

$$0^\circ\text{C} = 273.15 \text{ K}$$

Equations

$$PV = nRT$$

$$\frac{P_1 V_1}{n_1 T_1} = \frac{P_2 V_2}{n_2 T_2} = \frac{\text{rate}_1}{\text{rate}_2} = \sqrt{\frac{\text{molar mass}_2}{\text{molar mass}_1}}$$

$$q = mc_s \Delta T$$

PERIODIC TABLE OF THE ELEMENTS

PERIODIC TABLE OF THE ELEMENTS																		8A 18																									
1A 1																	2 18																										
1 H 1.008	2A 2															3A 13	4A 14	5A 15	6A 16	7A 17	8A 18																						
3 Li 6.941	4 Be 9.012															5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18																						
11 Na 22.99	12 Mg 24.31	13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95										19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.88	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.39	31 Ga 69.72	32 Ge 72.59	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80									
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc (99)	44 Ru 101.1	45 Rh 102.9	46 Pd 106.4	47 Ag 107.9	48 Cd 112.4	49 In 114.8	50 Sn 118.7	51 Sb 121.8	52 Te 127.6	53 I 126.9	54 Xe 131.3										55 Cs 132.9	56 Ba 137.3	57 La 138.9	58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm (145)	62 Sm 150.4	63 Eu 152.0	64 Gd 157.2	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0
87 Fr (223)	88 Ra 226.0	89 Ac 227.0	90 Th (232)	91 Pa (231)	92 U 238.0	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (260)	104 Rf (261)	105 Db (262)	106 Sg (266)	107 Bh (264)	108 Hs (277)	109 Mt (268)	110 Ds (281)	111 Rg (272)	112 Cn (285)	113 Nh (284)	114 Fl (289)	115 Mc (288)	116 Lv (291)	117 Ts (294)	118 Og (294)												
Lanthanides																																											
Actinides																																											

Lanthanides

58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu
140.1	140.9	144.2	(145)	150.4	152.0	157.2	158.9	162.5	164.9	167.3	168.9	173.0	175.0

Actinides

90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr
232.0	231.0	238.0	237.0	(244)	(243)	(247)	(247)	(251)	(252)	(257)	(258)	(259)	(260)

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